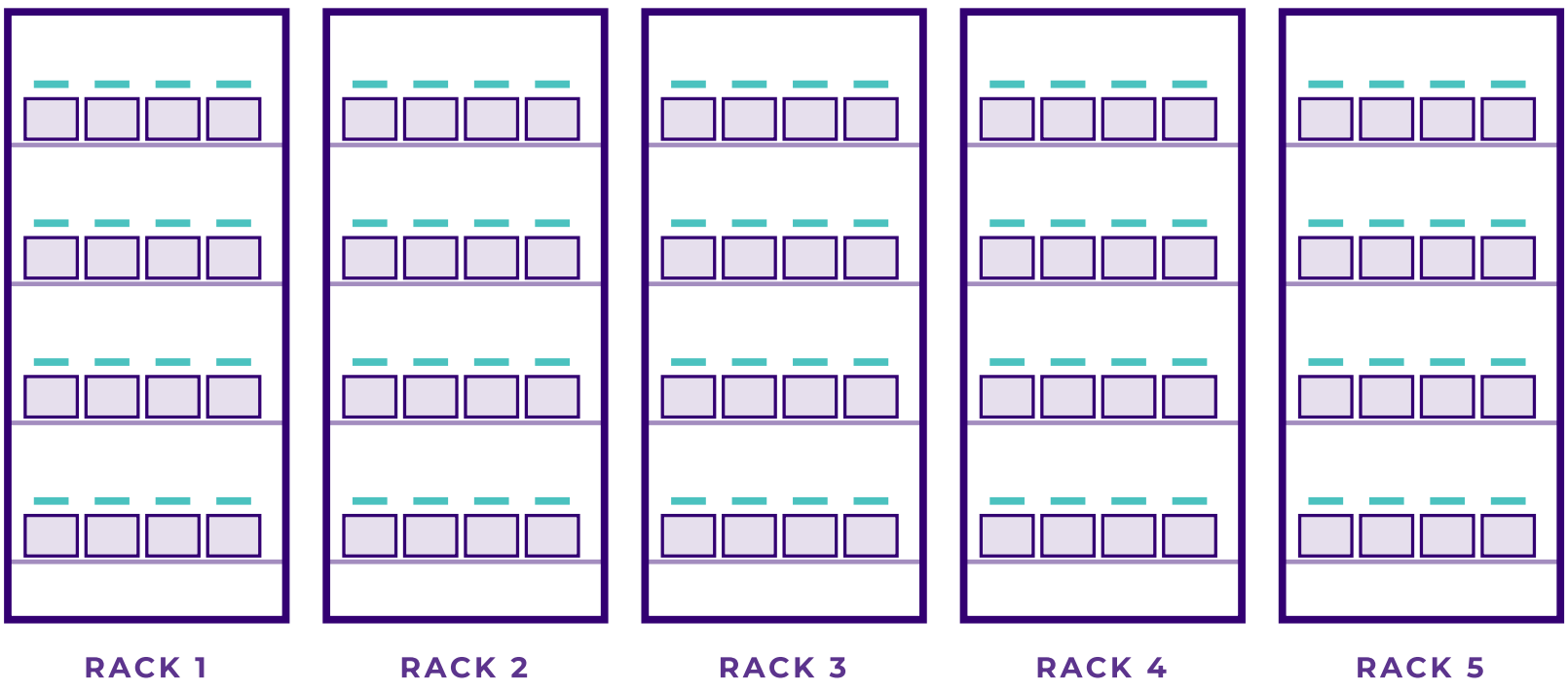
01

The microgreen process

One tray, ten to fourteen days · five stages, on the wall

Five stages, eighteen steps, ten to fourteen days. One stage to a sheet, on the wall where the work happens, in the order it happens.

80	200–300 g	136 g	10–14	5
TRAY PLACES	THE MODEL, A TRAY	THE MEASURED AVERAGE	DAYS TO CUT	STAGES



Five racks, four tiers, four trays to a tier: eighty places. One tray in and one tray out every day, so the room turns over once every twelve to fifteen days.

One tray in, one tray out. Eighty places divided by a twelve-day cycle is six or seven trays a day; by a fifteen-day cycle, five. The growth model runs two cycles a month — a hundred and sixty trays. The sow log is what settles it.

THE FIVE STAGES

1. **Set up and sow** — day 0, one tray at a time
2. **Blackout** — days 1–3, stacked, damp, dark
3. **Into the light** — day 4 until the cut
4. **Cut** — day 10–14, sharp, clean, weighed
5. **Pack and send** — same day, cold to the door

THE MACHINE DOES THREE THINGS

Lights 18 hours on, 6 off. **Pump** ten minutes at 08:00 and again at 16:00. **Fans** ten minutes in every thirty.

The room runs at 18–24 °C and 40–60 % humidity. Everything else on these eight sheets is hands.

02

The cycle, one tray

Day 0 to day 14 · and what a tray is worth in grams



One tray from sow to cut. The clock starts when the seed goes down, not when it comes into the light.

THE DAY, IN FIVE JOBS

JOB	HOW MANY	WHEN
Sow	**5–6 trays**	every day, one in one out
Stack, weight, mist	**the trays sown 1–3 days ago**	twice or three times a day
Move to the light	**the trays that have finished blackout**	every morning
Cut, weigh, pack	**the trays that are ready**	every morning, into the cold
Clear and clean	**every tray cut that day**	straight after the cut

One person, every day, five things. A tray moves one step further down the line each day, so the same hands sow, mist, lift, cut and pack without ever doubling back.

WHAT A TRAY ACTUALLY GIVES

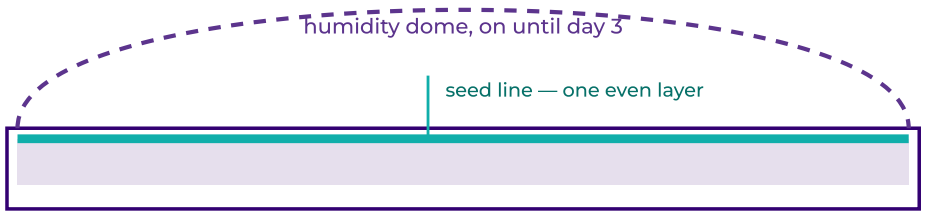
GREEN	GRAMS A TRAY
Mustard (Red Rain)	**201 g**
Hon Tsai Tai	**181 g**
Mizuna	**176 g**
Pac Choi	**170 g**
Basil	**165 g**
Persian cress	**164 g**
Sorrel	**163 g**
Kale (Russian)	**155 g**
Arugula	**148 g**
Kohlrabi (Purple)	**126 g**
Radish (Daikon)	**122 g**
Chinese cabbage (Kogane)	**120 g**
Kale (Toscano)	**118 g**
Tatsoi	**115 g**
Shungiku	**114 g**
Red cabbage	**106 g**
Ruby red chard	**71 g**
Cilantro	**47 g**

Measured at Penn State over 14 to 28 cycles a tray, ten to fifteen grams of seed, two weeks, no supplemental light: **136 g average, 47 to 201 g**. Bar marked in cyan is over 150 g. The growth model carries 200–300 g a tray — that is the number to beat, not the number to plan on until the room has weighed fifty trays.

Set up and sow

01

Day 0 · one tray at a time



10 × 20 in tray · medium level and firmed · seed broadcast to the edges

seed per tray, by variety: 3 g sorrel · 15–20 g brassica · 30–35 g radish · 200–275 g pea

01 Clean the tray

Wash it, sanitise with food-safe sanitiser, rinse, dry it upside down. A tray that still smells of the last crop is the next crop lost.

Watch · A dirty tray fails quietly — the mould shows on day four, not on day nought.

02 Fill the medium

Coco coir or a hemp mat, pre-moistened before it goes in. Level it to the rim, then firm it flat with the press.

Watch · Firm, not compacted. A hard tray roots on the surface instead of into it.

03 Weigh the seed

Weigh by variety, never by eye: 15–20 g for a brassica, 30–35 g radish, 200–275 g pea, 450 g wheatgrass. Every rate is on the variety sheet.

Watch · Seed by eye is how one tray comes up thin and the next one moulds.

04 Sow to the edges

Edges first, then the middle, so the last of the seed lands where the hand is not yet tired. Even coverage is the whole game.

Watch · A bare strip at the edge is a bare strip in the punnet.

05 Mist, dome, tag

One light mist, dome on, tag on the tray: variety, sow date, tray count, section. Then into the dark stack.

Watch · An untagged tray cannot be traced to a sow date or to a client. Nothing leaves the room untagged.

> One tray, one tag, one sow date. The tag is the only thing that survives all the way to the punnet.

Blackout

02

Days 1–3 · stacked, damp, dark



every tray in the stack carries its own tag — the stack is not a place to lose one

06 Stack and weight

Stack four high with a weight on top, or one dome per tray. The weight presses the roots down and pushes the stems up.

Watch · No weight and the crop comes up wearing seed coats on every leaf.

07 Keep it damp, not wet

Lift the stack and mist two or three times a day. Damp to the back of the hand, never standing water.

Watch · Standing water plus warmth is damping-off. It shows as a bald patch, never as a warning.

08 Check it even

By day three the tray is an even mat of pale stems. Patchy now is patchy at the cut.

Watch · Slow varieties are not three-day trays. Beet, chard, basil, coriander and chives run longer — read the variety sheet.

WHEN THE STACK GOES WRONG

Three signs, and what each one is telling you

A bald patch on one tray · Seed went on thin, or the mist missed the middle. It will not fill in — cut around it and log the tray.

Pale, stretched stems · Too warm and too crowded. It still grows and it still sells, but it is a leggy tray — cut it early and note the temperature.

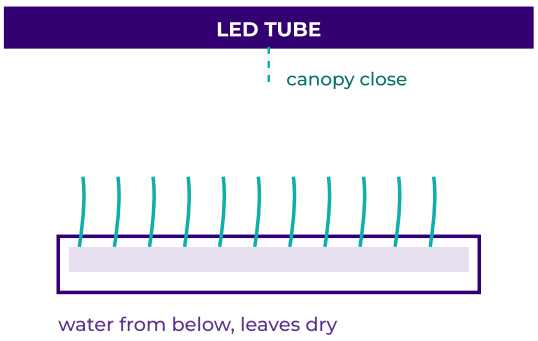
Grey fuzz, or a sour smell · Too wet, too warm, or no air. Out of the room in a bag, sealed, and do not open it to look. Wipe the level it stood on.

> Three days of dark, then light. Not five, not one — and not the same three days for every variety.

Into the light

03

Day 4 until cut · 12–16 hours a day under LED



THE MACHINE

lights
18 h on · 6 h off

pump
10 min at 08:00 and 16:00

fans
10 min in every 30

09 Move it under the light

Off the stack, under the tubes, canopy close. Twelve to sixteen hours a day colours the leaf without stretching the stem.

Watch · The machine runs the lights 18 on, 6 off. The protocol says 12–16. Pick one, write it on the wall and hold it — do not split the difference.

10 Water from below

Sub-irrigation keeps the leaves dry, and dry leaves are the whole trick. Let the tray drink, then pour off what is left.

Watch · Water on the canopy at this stage is a mould tray by Friday.

11 Mist less as it closes

Two or three times a day at first, less as the canopy fills. The room sits at 40–60 % humidity — let the room do half the work.

Watch · A closed canopy holds its own moisture. Keep misting it and it rots from the middle out.

12 The morning five

Temperature, humidity, air, water, and anything growing that is not a microgreen. Written down, every morning, before anything else.

Watch · 18–24 °C and 40–60 % humidity. A still room in a warm week is a mould week.

13 Read the variety, not the calendar

Radish is cut at day 5 or 6. Basil takes 20 to 25. The tray tells you: full colour, first true leaf just showing, stem firm.

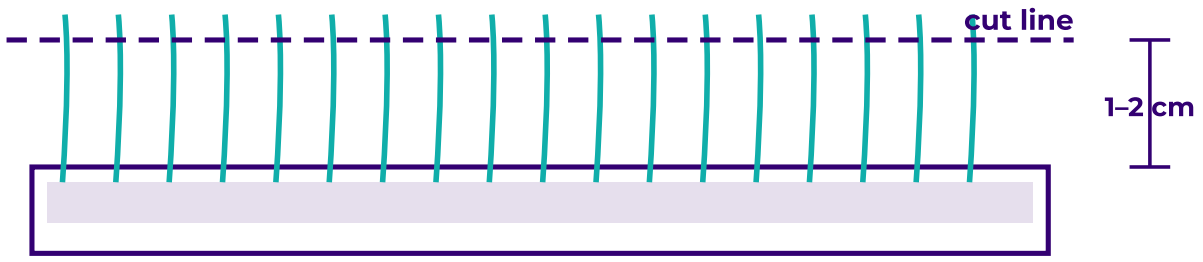
Watch · Past the first true leaf most varieties lose the flavour you are charging for.

> Light makes the colour, air makes the stem, and the variety decides the day.

Cut

04

Day 10–14 · sharp, clean, weighed



one clean pass, sharp blade — a torn stem is a punnet that goes limp early

14 Cut clean, one pass

Sharp scissors or a knife, 1–2 cm above the medium, one clean pass. Do not tear it and do not pull it.

Watch · A torn stem bleeds and the punnet goes limp a day early. Change the blade, not the pressure.

15 Weigh every tray

Grams on the sheet, tray by tray. The model says 200–300 g a tray; the research average is 136 g. The room's own number is the only one that pays.

Watch · An unweighed tray hides a thin crop until the invoice.

16 Straight into the cold

Into the cold room at 5 °C, covered, in the same hour it was cut. Not a warm tray on a bench, not even for a minute.

Watch · Two hours on a bench is two days off the shelf life.

A THIN TRAY, AND WHY

Four things to check before you blame the seed

The seed was weighed · Grams on the scale, not a scoop. Half a tray of seed is half a tray of greens, every time.

The light hours held · Twelve to sixteen, or the machine's eighteen — one setting, held for the whole cycle, not changed mid-tray.

The blackout was long enough · Variety sheet, not habit. Beet, coriander and chives want days more than a brassica does.

The water came from below · A tray that dried at the roots on day five never catches up. Weigh it anyway and write the number down.

> Cut height is the yield. One centimetre higher is a punnet that looks full and weighs light.

Pack and send

05

Same day · cold from the bench to the door



under two hours from the cold room to the client; Tuesday and Friday harvests go out the same day

5–7 days in a fridge. One day in a warm car. The cold chain is the product.

17 Pack, weigh, label

100 g clear clamshell for retail; 250 g or 500 g food-grade bag for wholesale. Variety, weight, harvest date and batch code on every one.

Watch · The batch code is the same code that was on the tray. If it is not, the trail stops at the pack bench.

18 Cold to the door

Cold room to client in under two hours, insulated bag, ice pack. Tuesday and Friday harvests leave the same day.

Watch · An hour in a warm car undoes the cold room.

CLEAR AND CLEAN, THE SAME AFTERNOON

A tray that is cut is a tray that is washed before the room closes

The spent tray goes out · Medium and roots to compost, mat to the bin, and out of the room — never beside the rack.

The level is wiped · Shelf wiped, floor swept, and anything mouldy out of the room the same hour it was found.

The tray is washed · Washed, sanitised, rinsed, dried upside down, and back on the rack in its place.

The room is shut down properly · Lights off, fans on their cycle, tank checked, door closed. The next shift starts from a clean room or it starts behind.

> Five to seven days in a fridge. One day in a warm car.

The variety sheet

08

Twenty-two greens · rate, soak, blackout, days

Seed rate is grams of dry seed for one 10 × 20 inch tray. Soak is before sowing. Blackout is days in the dark, stacked or domed. Days is sow to cut. Where a variety is not on this sheet it has not been grown here yet — write its line in by hand and the sheet stays true.

VARIETY	SEED A TRAY	SOAK	BLACKOUT	DAYS	NOTE
Fast · cut in 5–12 days					
Radish (Daikon, Sango)	**30–35 g**	none	**2–3 d**	**5–12**	Peak crunch at day 5–6
Kale (Vates, Red Russian)	**15–20 g**	none	**2–3 d**	**6–10**	Base green for a salad
Kohlrabi (Purple Vienna)	**15–20 g**	none	**2–3 d**	**6–10**	Purple stems, cheap seed
Mustard, mizuna	**10–15 g**	none	**3–5 d**	**8–12**	Spicy; runs cooler
Broccoli	**15–20 g**	none	**2–3 d**	**8–12**	The easiest tray in the room
Cabbage, pak choi	**15–20 g**	none	**2–3 d**	**8–12**	Purple pak choi needs less light
Cress, curled	**20 g**	none	**4–5 d**	**8–12**	Mucilaginous — mist light, 3× a day
Chia	**10–15 g**	none	**3–5 d**	**8–12**	Mucilaginous — sow thin
Buckwheat	**100 g**	4–8 h	**3–5 d**	**8–12**	Rinse well, husks must come off
Medium · cut in 10–14 days					
Beetroot (Bulls Blood, Detroit)	**20–30 g**	4–8 h	**4–5 d**	**10–14**	Harvest low to keep the red stem
Swiss chard (all stems)	**20–30 g**	4–8 h	**4–5 d**	**10–14**	Poor in hydroponics — soil tray
Amaranth (Red Garnet)	**15 g**	none	**5–6 d**	**12–14**	Wants warmth; burns in direct light
Lettuce	**15–20 g**	none	**2–3 d**	**10–12**	Petite; colour follows the light
Sorrel, red-veined	**3 g**	none	**3–5 d**	**10–12**	Expensive seed — sow sparse
Fennel	**20–30 g**	none	**3–4 d + 2–3**	**16–20**	Husks cling; grow it long enough
Parsley	**15–20 g**	none	**4–5 d**	**16+**	Slow to germinate, soil only
Slow · cut in 16–25 days					
Basil (Genovese, Dark Opal)	**10 g**	none	**4–7 d**	**20–25**	Mucilaginous; keep it damp
Coriander	**30–40 g**	2–4 h	**stack 7 d**	**21–28**	Split seed germinates faster
Chives, spring onion	**40–50 g**	none	**4 d + 2–3**	**21**	Slow; can give a second cut
Heavy seed · soak first					
Pea shoots	**200–275 g**	6–12 h	**stack 3–5 d**	**8–12**	Pre-sprout 12–24 h; 12 days max
Sunflower (black oil)	**150 g**	6–12 h	**2–3 + 2–3**	**8–12**	Pre-sprout 12–24 h; soil only
Wheatgrass	**450 g**	4–8 h	**2 d**	**8–10**	Juicing only; mould-prone, keep cool

Rates, soaks and blackout days are trade practice from the Bootstrap Farmer cheat sheet, not yet measured in this room — treat them as the starting point and overwrite them with what the trays do. Seed rate sanity-checks against Penn State Extension, who sow **10–15 g** a tray for the brassicas and get 47–201 g back. Heavy seed — pea, sunflower, wheatgrass — is weighed, not counted: it is ten to twenty times the brassica rate.